SDI Training: Adapted Plan for ICT Users

GeoNode, KoboToolbox & Climate Risk Database

Non-Programmer Edition — Computer-Literate Professionals

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Duration: 4 Days
Target: ICT officers, GIS technicians, data officers, M&E staff
Prerequisites: Comfortable with computers, file management, web browsers
Platform: crd.resilienceacademy.ac.tz

Philosophy: What Changes for Non-Programmers?

The original training plan assumes participants who can write terminal commands, configure YAML files, and debug Docker containers. For ICT users with computer background (but no programming experience), we restructure around these principles:

1. **GUI-first, terminal-second.** Use web interfaces and desktop applications as the primary interaction. Terminal commands are introduced as *copy-paste recipes*, not as skills to master.
2. **"Why" before "How".** Spend more time on concepts, use cases, and demonstrations before hands-on. Participants must understand *what* CRD/KoboToolbox does for their institution before learning *how* it works internally.
3. **Guided walkthroughs, not independent exploration.** Every practical has an instructor doing it live on the projector while participants follow step-by-step. No "figure it out" exercises.
4. **Reduce scope, increase depth on usable skills.** Cut bare-metal Ubuntu installation, advanced Docker debugging, and Python scripting. Add more time for: using the CRD web interface, uploading/managing data, building KoboToolbox forms, and consuming data in QGIS.
5. **Real outcomes each day.** Every day ends with something the participant can show: a dataset on the map, a working survey form, a QGIS project with live CRD layers.

Audience Profile

Characteristic	Description
Job roles	ICT officers, GIS technicians, data/M&E officers, planning officers, environmental officers
Computer skills	Comfortable with Windows/Mac, file management, web browsers, Excel/Word, email
Programming	None or minimal (may have seen HTML or Excel formulas)
GIS experience	May have used Google Maps/Earth; some may know QGIS basics
Linux/Terminal	Little to no experience
What they need	Ability to *use, manage, and populate* CRD and KoboToolbox — not build from source

Key Differences from Original Plan

Topic	Original (Developer)	ICT Users (This Plan)
Installation	Terminal commands, Docker build, bare-metal setup	Pre-built Docker script with one-click installer; instructor handles setup; participants observe and verify
Configuration	Edit .env files, local_settings.py	Web-based admin panel; instructor-guided .env editing with copy-paste
Data management	curl API, Python upload scripts	Web interface upload, drag-and-drop, QGIS import
KoboToolbox forms	XLSForm in Excel + API	Form Builder GUI (visual drag-and-drop)
Integration	Python scripts (kobo_to_crd.py)	Export CSV from Kobo → open in QGIS → save as GeoJSON → upload to CRD via web
API usage	Python requests, Postman, REST	Browser-based API explorer; Postman as visual tool; no coding
Linux commands	Full session (3h practical)	1h guided orientation; focus on 10 essential commands as recipes
Git	Clone, branch, commit workflow	Concept only; instructor handles git clone; participants don't need Git
Docker	Build, debug, compose YAML	Concept only ("magic box" analogy); participants run pre-made scripts
QGIS	Brief WMS/WFS connection	Extended: load layers, style, query, print layout, export
Time on concepts	30% lectures, 70% hands-on	40% lectures/demos, 60% guided hands-on

Day 1: Understanding SDI & Getting Started

Morning — Lectures & Demonstrations (4 hours)

Session 1.1 — What is Spatial Data & SDI? (1.5 hours)

- What is spatial data? (definition, examples from Tanzania)
- Vector vs raster data — with visual examples from CRD
- Coordinate systems explained simply (GPS coordinates = lat/lon)
- What is an SDI? The five pillars (data, metadata, services, standards, people)
- Why SDI matters for your institution (climate risk, planning, reporting)
- OGC standards explained with analogies (WMS = photo of map, WFS = raw data file)
- **Live demo:** Browse crd.resilienceacademy.ac.tz together
- **Exercise:** Participants search for Tanga and Mwanza datasets on the live CRD

Session 1.2 — The Technology Behind CRD (1 hour)

- GeoNode explained: "WordPress for maps" analogy
- Architecture overview (simplified): Browser → Web Server → Map Server → Database
- Docker explained: "shipping container" analogy — your app runs the same everywhere
- No need to understand Docker internals; just know it exists and what it does
- Software overview: what each tool does and *when participants will use it*
- Hardware requirements discussion

Session 1.3 — Introduction to the Linux Environment (1.5 hours)

- Why Linux? (all SDI software runs on it)
- Opening a terminal (Ubuntu desktop or WSL2 on Windows)
- **10 essential commands** taught as copy-paste recipes:
  1. `pwd` — where am I?
  2. `ls -la` — what files are here?
  3. `cd folder` — go to a folder
  4. `mkdir name` — create a folder
  5. `cp / mv / rm` — copy, move, delete
  6. `cat file.txt` — view a file
  7. `nano file.txt` — edit a file (with guided save/exit)
  8. `sudo command` — run as administrator
  9. `df -h` — check disk space
  10. `curl URL` — fetch data from the web
- **Guided exercise:** Navigate to a folder, create a file, view it, delete it
- **Guided exercise:** Use curl to fetch CRD API response (copy-paste the command)

Afternoon — Guided Setup (3 hours)

Session 1.4 — Setting Up the Environment (3 hours)

- **Instructor-led, step-by-step** — projector + participants follow
- Install Docker Desktop (Windows/Mac) or Docker Engine (Ubuntu) — using installer wizard
- Verify: `docker run hello-world` (copy-paste)
- Install QGIS from qgis.org (graphical installer)
- Install VS Code (optional, for those who want to see config files)
- **Checkpoint:** Docker works, QGIS opens, browser accesses crd.resilienceacademy.ac.tz
- Troubleshooting time with peer support

Day 1 Outcome: Every participant understands what SDI is, can navigate the live CRD platform, knows 10 Linux commands, and has Docker + QGIS installed.

Day 2: CRD Installation & Data Management

Morning — CRD Installation (4 hours)

Session 2.1 — CRD Architecture & Live Walkthrough (1 hour)

- Detailed walkthrough of crd.resilienceacademy.ac.tz:
  - Catalogue: browse, search, filter datasets
  - Dataset page: metadata, map view, download options
  - Developer page: WMS/WFS endpoint URLs
  - Admin panel: user management, site settings
- Participants follow along on their own browsers
- Q&A: "How would this help your institution?"

Session 2.2 — Installing CRD with Docker (3 hours)

- **Instructor does every step live; participants copy-paste commands**
- Provide a **command sheet** (printed or PDF) with every command numbered
- Steps (all copy-paste):
  1. `git clone https://github.com/GeoNode/geonode-project.git my-crd`
  2. `cd my-crd`
  3. `cp .env.sample .env`
  4. Instructor explains 3 key .env settings (SITEURL, password, secret key) — participants edit with `nano`
  5. `docker compose build` — wait (explain what is happening)
  6. `docker compose up -d` — wait
  7. `docker compose logs -f` — watch startup
  8. Create admin account (copy-paste command)
  9. Open `http://localhost` in browser
- **Checkpoint:** CRD loads in browser; admin login works
- 30 min troubleshooting buffer

Afternoon — Using CRD (3 hours)

Session 2.3 — CRD Customisation via Web Admin (1.5 hours)

- All done through the **web browser admin panel** — no code editing
- Change site name and domain
- Upload a logo
- Configure user registration (open vs invite-only)
- Set default map centre (Tanga or Mwanza coordinates)
- Create user accounts and set permissions

- **Checkpoint:** Customised CRD with institution name and logo

Session 2.4 — Uploading & Managing Data Layers (1.5 hours)

- **Via CRD web interface** (Data → Upload Layer):
  - Upload a Shapefile (mwanza_businesses)
  - Upload a GeoJSON file
  - Upload a GeoTIFF raster
- Set metadata: title, abstract, keywords, category
- Set permissions: public view, group edit
- View layers on the map
- **QGIS connection:** Add CRD layers via WMS/WFS (Layer → Add WMS/WFS Layer)
- **Checkpoint:** At least 2 layers uploaded and visible on map + loaded in QGIS

Day 2 Outcome: Every participant has a running local CRD with custom branding, uploaded data layers visible on the map, and those same layers loaded in QGIS.

Day 3: KoboToolbox for Data Collection

Morning — KoboToolbox Installation & Forms (4 hours)

Session 3.1 — KoboToolbox Overview & Use Cases (1 hour)

- What is KoboToolbox? Mobile data collection platform
- RA/GIZ context: Tanga flood mapping with 16 alumni
- Cloud vs self-hosted: why self-host? (data sovereignty, institutional requirements)
- KoboCollect app: offline GPS data collection on Android
- Live demo: show a deployed form, submit test data, view results

Session 3.2 — Installing KoboToolbox (2 hours)

- Same approach as CRD: **instructor-led, copy-paste commands**
- Printed command sheet provided
- Steps:
  1. Clone kobo-install repository
  2. Run setup wizard: `python3 run.py` (answer prompts together)
  3. Start services and wait for containers
  4. Create admin account
  5. Access `http://localhost` — KoboToolbox login page
- **Checkpoint:** KoboToolbox running; admin login works

Session 3.3 — Designing a Survey Form (1 hour)

- **Using the visual Form Builder** (drag-and-drop in browser — no XLSForm)
- Build a Tanga flood survey form:
  - GPS location (geopoint)
  - Flood depth (decimal number)
  - Date of observation
  - Ward name (select from list)
  - Photo attachment
  - Number of affected households
  - Infrastructure damage type (select multiple)
- Deploy the form
- Submit 3 test entries using Tanga coordinates
- View submissions in the data table
- **Checkpoint:** Form deployed, test submissions visible

Afternoon — Management & Integration (3 hours)

Session 3.4 — KoboToolbox Administration (1.5 hours)

- User management via admin panel (create accounts, assign permissions)
- Check system health: `docker compose ps` (copy-paste)
- Backup: instructor demonstrates; participants follow
- Export data: CSV, Excel, GeoJSON downloads from the Kobo web interface

Session 3.5 — Kobo to CRD: The GUI Way (1.5 hours)

- **No Python scripts required.** Instead:
  1. Export survey data from KoboToolbox as **CSV** (Downloads section)
  2. Open CSV in **QGIS**: Layer → Add Delimited Text Layer → set X/Y columns
  3. Verify points appear on the map at correct Tanga locations
  4. Export from QGIS as **GeoJSON**: Right-click layer → Export → Save As GeoJSON
  5. Upload GeoJSON to **CRD web interface**: Data → Upload Layer
  6. Set metadata and permissions
  7. View flood survey points on CRD map overlaid on Tanga imagery
- **Checkpoint:** Tanga flood survey data visible and clickable in CRD

Day 3 Outcome: KoboToolbox running, flood survey form designed and tested, survey data exported and uploaded to CRD via QGIS — no coding required.

Day 4: Using CRD Data & Wrap-Up

Morning — QGIS with CRD Data (3.5 hours)

Session 4.1 — QGIS Deep Dive with CRD Layers (3.5 hours)

- **Extended QGIS session** (replaces Leaflet JavaScript coding from original plan)
- Connect QGIS to live CRD:
  - Add WMS layer: Layer → Add WMS/WMTS Layer → New Connection
  - URL: `https://crd.resilienceacademy.ac.tz/geoserver/ows`
  - Load mwanza_businesses, mwanza_education_facilities
  - Load Tanga_imagery_reduced as background
- Add WFS layer (vector download):
  - Layer → Add WFS Layer → same URL
  - Load mwanza_businesses as editable vector
- Style layers by attribute (categorised/graduated symbology)
- Open attribute table, filter, select features
- Use Sketcher/Sketcher: add labels, change colours
- **Print layout:** Create a professional map with title, legend, scale bar, north arrow
- Export map as PDF or PNG
- **Checkpoint:** Professional map exported with CRD data, legend, and layout

Afternoon — Exploring the API & Closing (3.5 hours)

Session 4.2 — CRD API for Non-Programmers (1.5 hours)

- **Browser-based API exploration** (no code):
  - Open: `crd.resilienceacademy.ac.tz/api/v2/datasets/`
  - Explain JSON structure visually
  - Search: `/api/v2/datasets/?search=mwanza`
  - Count datasets, find download links
- **Postman** as a visual API tool:
  - Install Postman (graphical app)
  - Create a GET request to CRD API
  - View formatted JSON response
  - Save requests as a collection
- Explain when/why APIs matter: automated reporting, dashboards, mobile apps

Session 4.3 — Bringing It All Together (1 hour)

- Recap exercise: each participant demonstrates their full workflow:
  1. Show their customised local CRD
  2. Show uploaded data layers on the map
  3. Show their KoboToolbox form and submissions
  4. Show their QGIS map layout with CRD data
- Group discussion: "How will you use this at your institution?"
- Next steps: deploy on a cloud server with a real domain name

Session 4.4 — Wrap-Up & Closing Ceremony (1 hour)

- Review of all 4 days
- Resources: docs.geonode.org, kobotoolbox.org/documentation, crd-userguide.readthedocs.io
- Feedback forms
- Certificate distribution
- Closing remarks

Day 4 Outcome: Professional QGIS map with CRD data, understanding of CRD API, and a complete demonstration of the full data workflow from collection (Kobo) to publication (CRD) to analysis (QGIS).

Summary: Original vs ICT Users Plan

Day	Original (Developers)	ICT Users (This Plan)
Day 1	4 lectures (SDI, software, servers, DevOps) + 3 practicals (environment setup, Linux, Git/Docker)	3 lectures (SDI concepts, technology overview, Linux basics) + 1 guided setup practical
Day 2	CRD architecture lecture + Docker install + Ubuntu install + customisation + data loading	CRD walkthrough + Docker install (guided) + web admin customisation + web data upload + QGIS connection
Day 3	Kobo lecture + install + XLSForm design + backend management + Python integration	Kobo overview + install (guided) + visual form builder + admin basics + GUI integration (CSV → QGIS → CRD)
Day 4	Leaflet JS web app + Python/GeoPandas + REST API dashboard	Extended QGIS session + browser API exploration + Postman + portfolio demo + closing

Materials Needed (Additional)

- **Command sheets** — printed A4 sheets with numbered copy-paste commands for each practical
- **Cheat cards** — laminated quick-reference cards for 10 Linux commands and QGIS shortcuts
- **Sample datasets** — pre-downloaded GeoJSON/Shapefile files on USB drives (backup for slow internet)
- **Pre-built Docker images** — cached on a local server to speed up `docker compose build`
- **QGIS project templates** — .qgz files pre-configured with CRD WMS/WFS connections
- **Video recordings** — screen recordings of each installation for participants to review later

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